

Microbiology - Emerging Blood Borne Pathogens

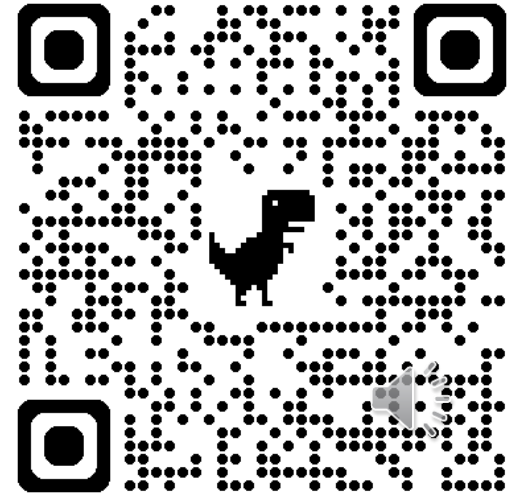
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Make.
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Innovate.

Reinvent the Future.

- Session Objectives:
 1. Review epidemiologic, virulence factors, and clinical presentations of emerging blood-borne pathogens
 2. Review diagnostic techniques and treatments for emerging blood-borne pathogens
 3. Compare and contrast the diseases caused by different species of single genuses, e.g., *Bartonella*, *Plasmodium*, and *Leishmania* spp.
- Connections
 - ScholarRx: [Protozoa and Ectoparasites](#)



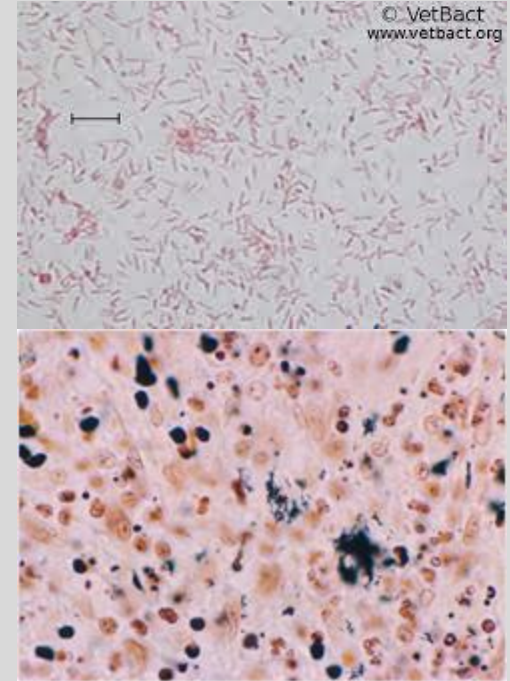
Key Concepts

- **Zoonosis:** human disease derived from an animal reservoir
 - e.g., Lyme disease from rodents, *Trypanosoma* from domesticated and wild animals
- **Vector:** a living organism, typically an arthropod, that transmits an infection from one host to another
 - e.g, mosquito, tick
- **Life cycle:** the phases of growth and transmission that results in maturity and reproduction, often requiring intermediate and definitive hosts
 - e.g., *Leishmania* life cycle through rodents, sand flies and humans
- **Infestation:** the presence of parasites or ectoparasites on the body surface
 - e.g., head lice (*Pediculus humanus capitis*) infestation



Bacteria: **Bartonella**

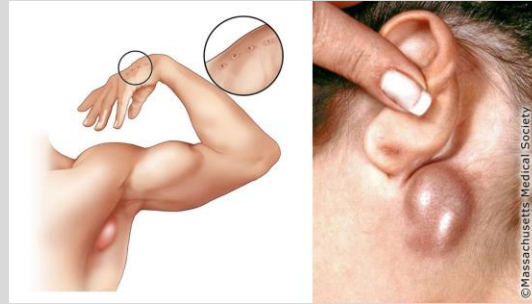
- *B. henselae*, *B. quintana*, *B. bacilliformis*
(many species cause human disease)
- **Pleiomorphic**, Gram-negative rod
- **Catalase/oxidase negative**
- Infect CD34+ cells (progenitors of endothelial cells) and erythrocytes, followed by dissemination to endovascular, lymphatic and other organs
- Culture: slow growth (2-6 weeks)
- Stain: **Warthin-Starry silver stain**



	<i>Bartonella henselae</i>	<i>Bartonella quintana</i>	<i>Bartonella bacilliformis</i>
Disease	Cat-scratch disease, bacillary angiomatosis, Bartonella peliosis 	Trench fever, bacillary angiomatosis 	Oroya (Carrión) fever (acute phase); verruga peruana (chronic phase) 
Geographic distribution; incidence	Worldwide; about 25,000 cases in U.S./y	Worldwide (poverty, overcrowding, experiencing homelessness)	High-altitude Andean valleys of Peru; also Ecuador, Columbia; outbreaks of thousands of cases
Transmission	Cat scratch or bite (cat flea)	Body louse feces	Sandfly bite
Incubation period	3-5 days	15-25 days	3-12 weeks
Clinical features	Red papule, vesicle or nodule at site of inoculation, regional lymphadenopathy, fever, malaise	Fever (quintan = every 5 days), rash, headache, back and leg pain, may be relapsing	Fever, weakness, myalgias, arthralgias; verruca peruana rash (Peruvian warts)
Complications	Endocarditis (culture-negative) , FUO, neuroretinitis Bartonella peliosis due to endothelial proliferation of liver and spleen	Endocarditis (culture-negative)	<i>B. bacilliformis</i> entry into and multiplication in erythrocytes causing membrane deformation and severe hemolytic anemia (acute phase)
Diagnosis	PCR, biopsy (stellate necrotizing granulomas), serology, blood culture	PCR, blood or tissue culture, serology	Blood culture, peripheral smear
Prognosis	Self-limited in immune competent, fatal if untreated in immune compromised	Good with prolonged treatment	Good with prolonged treatment; 90% mortality untreated
Treatment	Azithromycin or doxycycline	Trench fever: doxycycline + gentamicin Bacillary angiomatosis: erythromycin or doxycycline	Oroya fever (ciprofloxacin + ceftriaxone); verruga peruana: rifampin or streptomycin

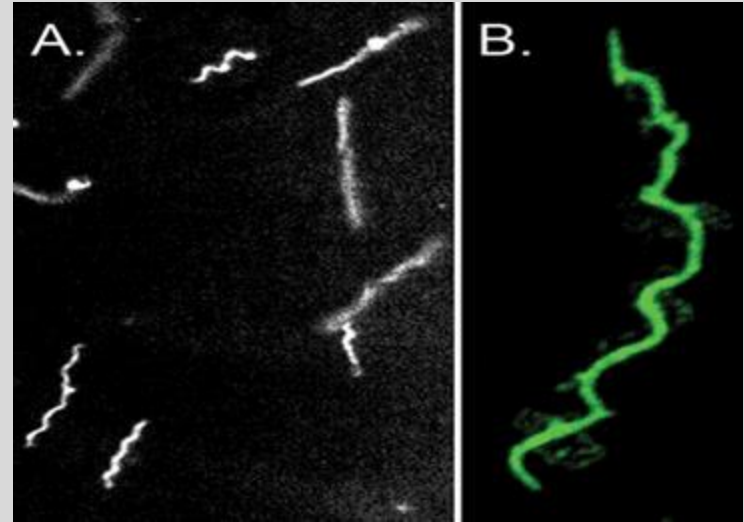
Bartonella Skin Lesions

- Cat-scratch papule occurs at inoculation site prior to regional lymphadenopathy
- Bacillary angiomatosis
 - *Bartonella henselae*, *B. quintana*
 - Host: immune suppressed, e.g., **AIDS CD4+ <100**
 - Pathogenesis: capillary proliferation
 - Diagnosis: **biopsy of lesion shows neutrophilic infiltrates**, plump endothelial cells (**epithelioid**) and basophilic aggregates of *Bartonella* organisms; PCR; blood culture (prolonged incubation)
 - Differential diagnosis: **Kaposi's** sarcoma (also seen in immune suppressed)
- Verruga peruana
 - Chronic, eruptive skin phase of Carrion's disease (*B. bacilliformis*)
 - Localized angioblastic hyperplasia (endothelial cell proliferation) forming blood-filled vascular nodules (verrugas)



Bacteria: *Borrelia*

- Class Spirochaete includes *Borrelia*, *Treponema*, *Leptospira* (although Gram-negative, they stain poorly and require special methods to visualize (e.g., Dark field [A], fluorescent [B])
- Corkscrew morphology, motile (flagella), fastidious growth
- Transmitted by ticks or lice; rodent and bird reservoir
- *B. burgdorferi* (Lyme disease)
 - At least 12 *Borrelia* species cause Lyme
- *B. miyamotoi*, *B. recurrentis*, *B. hermsii*, *B. parkeri* (relapsing fever)



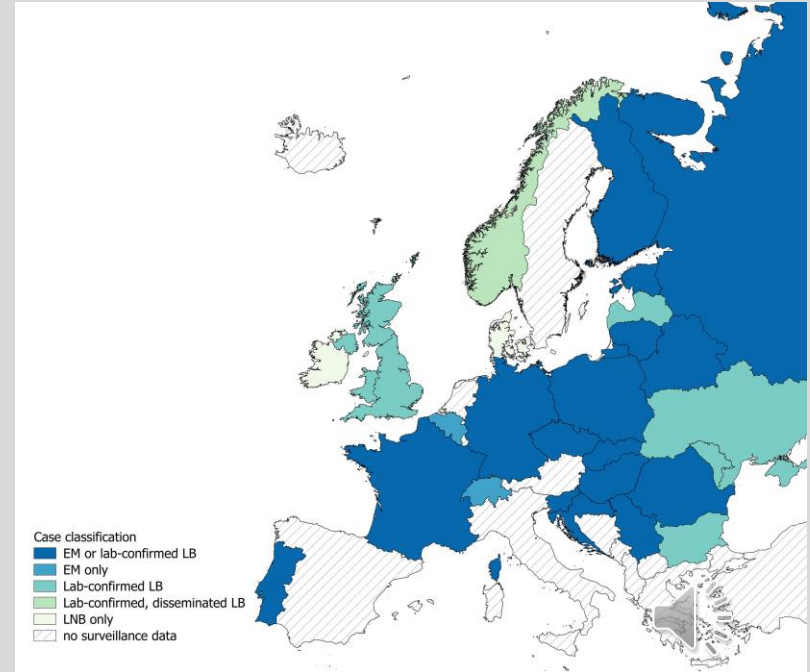
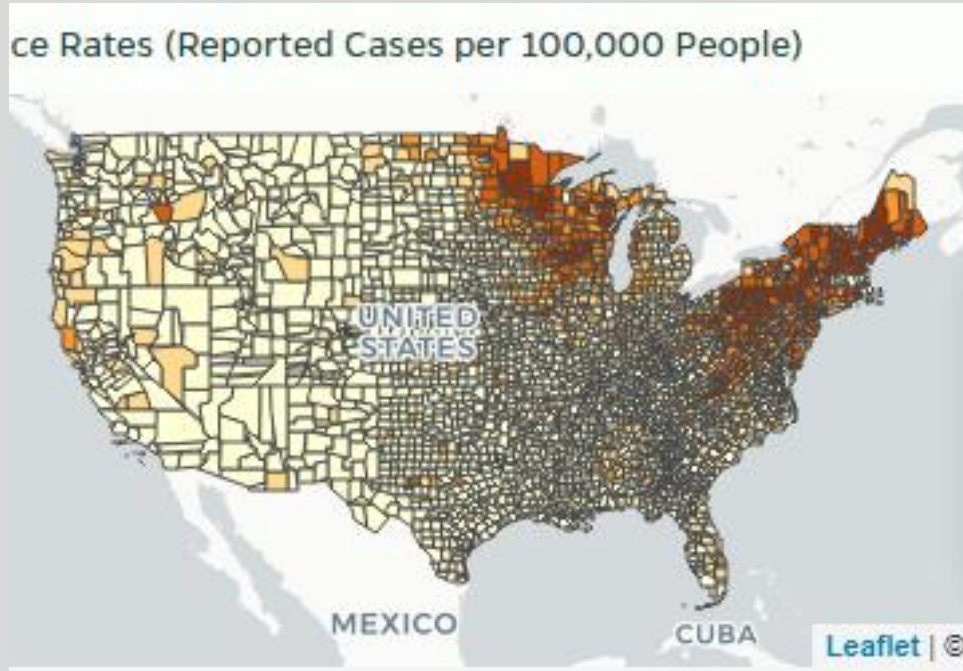
	<i>Borrelia burgdorferi</i>	<i>Borrelia miyamotoi</i>
Disease	Lyme disease	Hard tick relapsing fever
Geographic distribution, incidence	Northeast, central east coast and Midwest U.S. (enlarging), Europe U.S.: 500,000 cases/y (increasing, under reported), Europe >200,000/y	Northeastern, mid-Atlantic, and Upper Midwest U.S., California, Europe, Russia, Japan Disease is much less common; about 1-3% show evidence of infection in endemic areas
Transmission	Ixodes (deer) tick	Ixodes (deer) tick
Incubation period	3-30 days (as short as)1-3 days	12 – 16 days (3 days to 6 weeks)
Clinical features	Stage 1: Erythema migrans Stage 2: Early dissemination Stage 3: Late-stage	Fever (may or may not last several days separated by afebrile periods), chills, headache, myalgia, severe fatigue
Complications	Arthritis, meningitis, cardiac conduction block Co-infection (B. miyamotoi, Anaplasma, Babesia) Europe: acrodermatitis chronica atrophicans	Meningoencephalitis Co-infection
Diagnosis	Clinical (erythema migrans) 2-step serology (ELISA → Western blot) Wet mount dark-field	PCR Anti-glpQ antibody
Prognosis	Excellent with treatment; may result in prolonged sequelae	Excellent with treatment
Treatment	Doxycycline preferred; amoxicillin; ceftriaxone for CNS involvement	Doxycycline, amoxicillin, ceftriaxone



Lyme Geographic Distribution

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[Johns Hopkins Lyme Tracker](#)

Lyme Disease Stages

- Stage 1: Early localized infection
 - Erythema migrans (bullseye or target lesion)
 - Occurs 3-30 days from site of tick bite
 - Advancing Borrelia and lymphocyte/macrophage infiltration
- Stage 2: Early disseminated infection
 - Occurs days to months after stage 1
 - Transient bacteremia results in dissemination
 - Intermittent rash, migratory arthralgias, headache, facial (Bell's) palsy
 - Carditis: AV block, myopericarditis
 - Arthritis: non-symmetric, oligoarticular
- Stage 3: Late-stage infection
 - Occurs months to years after stage 1
 - Encephalopathy: altered mood, sleep, language; memory loss; brain fog
 - Acrodermatitis chronica atrophicans (Europe)



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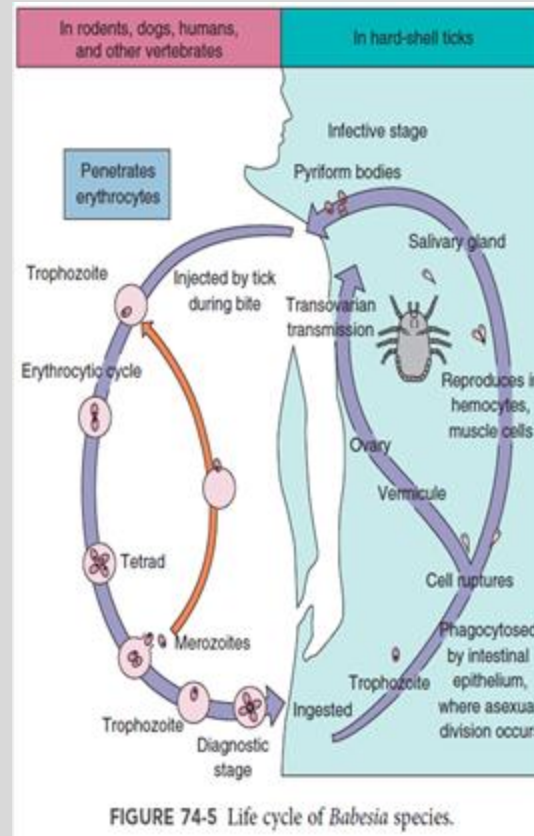


Protozoa: *Babesia microti* (Babesiosis)

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	<i>Babesia microti</i>
Disease	Babesiosis
Geographic distribution	NE, Midwest U.S. About 2000 cases/y
Transmission	Ixodes (deer) tick
Incubation period	1-4 weeks
Clinical features	Fever, chills, fatigue, myalgias, usually in summer
Complications	Hemolytic anemia, hepatosplenomegaly, thrombocytopenia Coinfection (Lyme, <i>Anaplasma</i>)
Diagnosis	Blood smear (Giemsa stain): Maltese cross , ring forms within RBC PCR
Prognosis	Good with treatment Disease worse in immunocompromised, elderly Asplenia: severe infection
Treatment	Moderate: atovaquone + azithromycin Severe: clindamycin + quinine



Maltese cross
(tetrad)

Ring form

Protozoa: *Plasmodium* (Malaria)



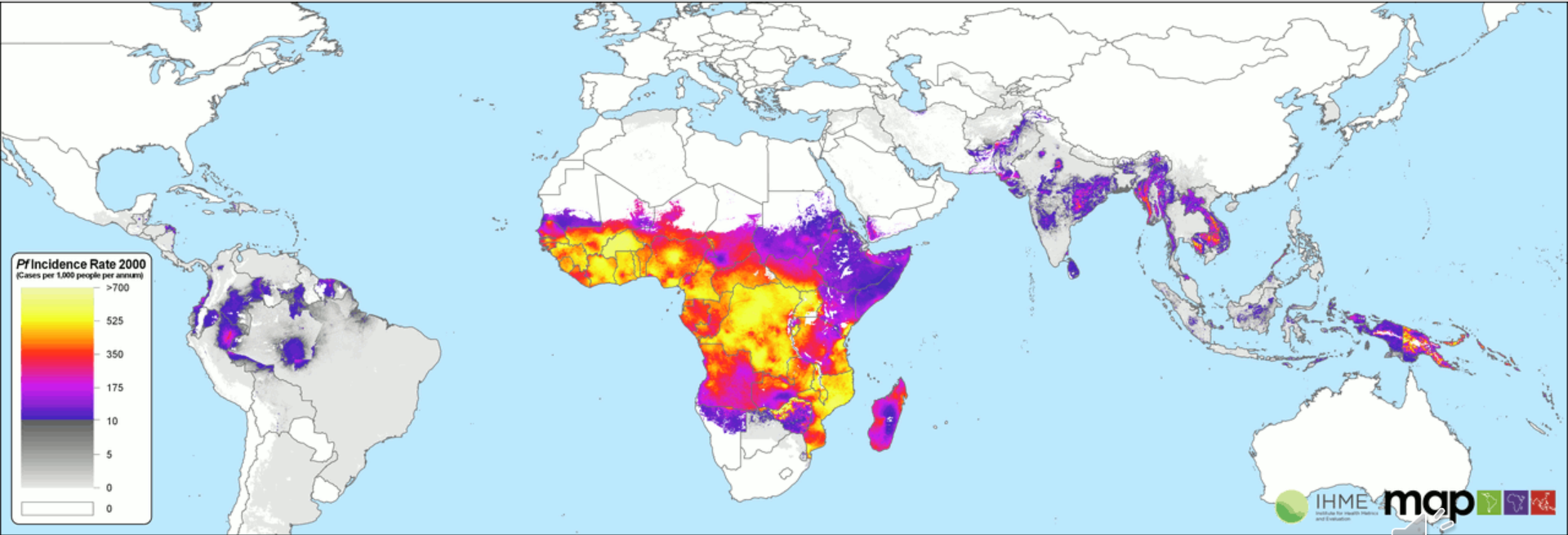
- Transmitted by *Anopheles* mosquito
- Severe, often fatal illness
- About 300 million infections, >600,000 deaths worldwide/y, mostly children and pregnant
- Entirely preventable and treatable, although drug resistance is increasing



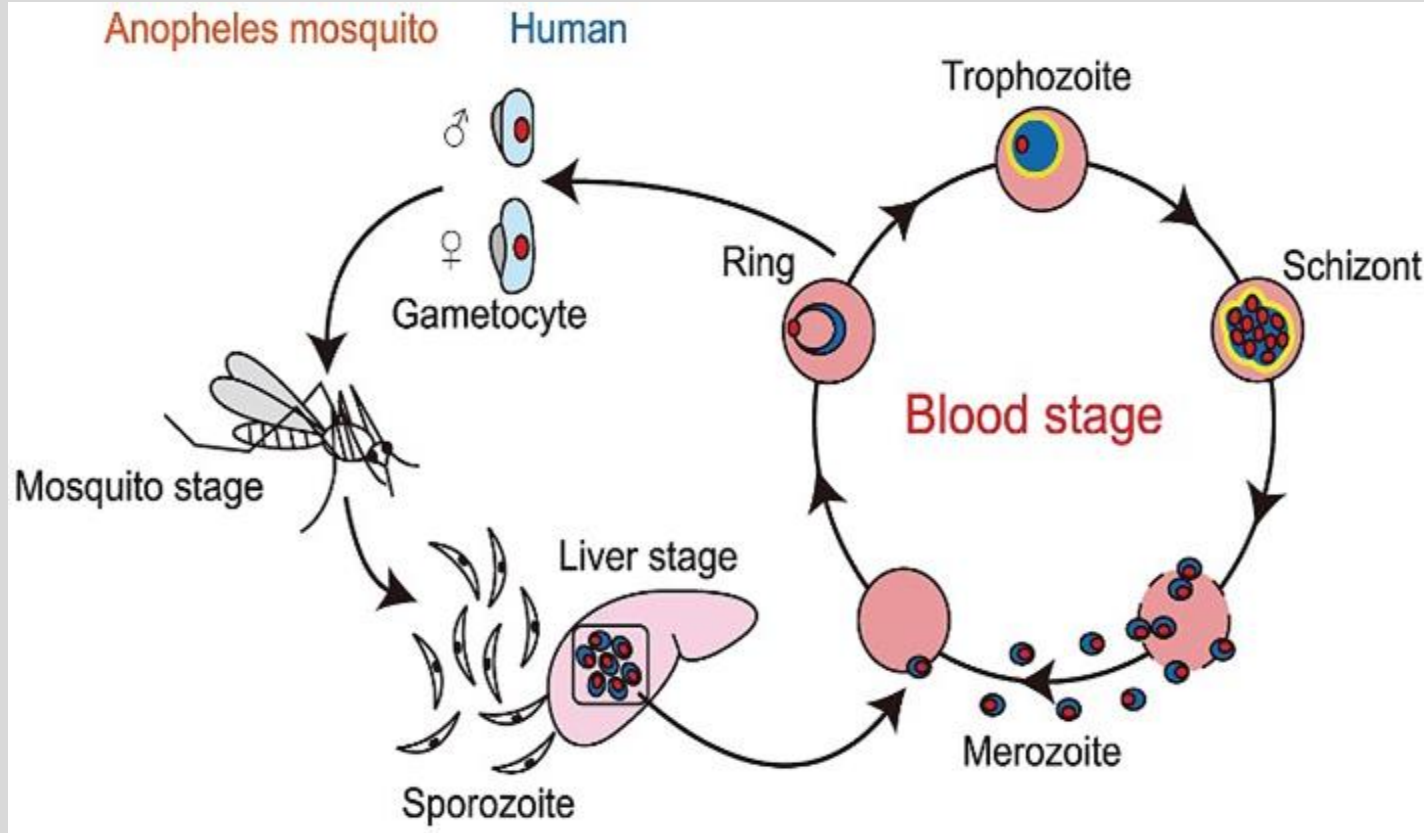
Malaria Geographic Distribution

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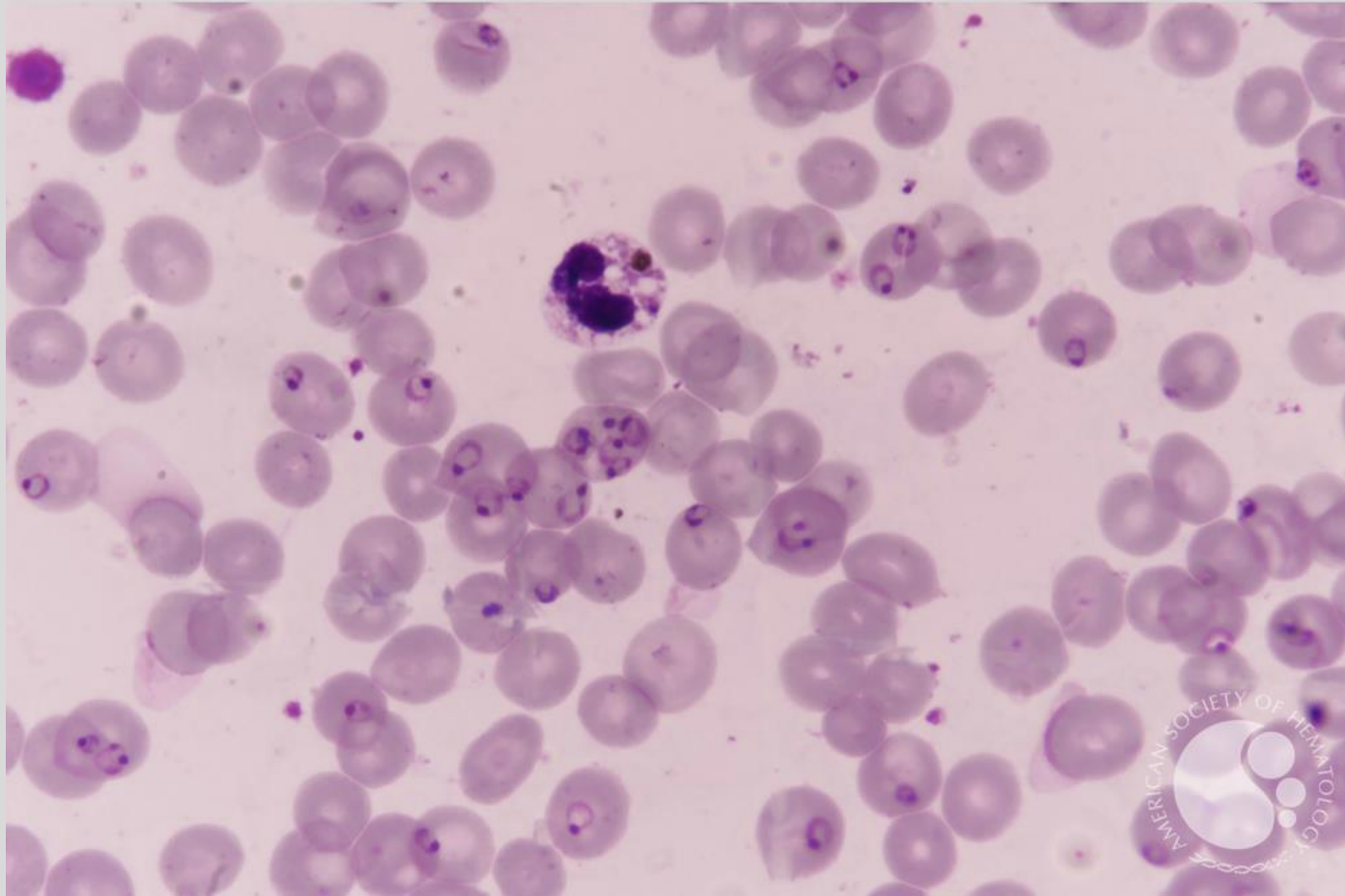


Plasmodium life cycle



	<i>Plasmodium falciparum</i>	<i>Plasmodium knowlesi</i>	<i>Plasmodium malariae</i>	<i>Plasmodium vivax</i>	<i>Plasmodium ovale</i>
Geographic distribution,	Sub-Saharan Africa, Papua New Guinea, SE Asia (Highest worldwide prevalence)	Forested regions of SE Asia, esp. Malaysian Borneo (rarest prevalence of 5 species)	Africa (including sub-Saharan), S. America, SE Asia (widest distribution)	Africa, S. America, SE Asia (dominant species outside sub-Saharan Africa)	Tropical Africa, S. America, SE Asia
RBC preference	None	None	Mature (senescent)	Young (larger)	Young (larger)
Incubation period	9-14 days	9-12 days	18-40 days	12-18 days	12-18 days
Fever pattern	Irregular or continuous	Quotidian (every 24 hours),	Quarten (every 3 days)	Tertian (every 48 hours)	Tertian (every 48 hours);
Other clinical features	Most severe form of malaria: Chills, headache, malaise, nausea, vomiting, diarrhea; hemolytic anemia, elevated LDH, hypoglycemia, acidosis	Headache, rigors, malaise, abdominal pain, SOB, productive cough; elevated transaminases, severe thrombocytopenia	Causes milder, subacute or chronic infection, mild anemia	“Benign malaria,” moderate anemia, thrombocytopenia, elevated LDH	“Benign malaria,” mild anemia and thrombocytopenia (rarely causes severe illness or death)
Complications	Blackwater fever: hemoglobinuria, acute tubular necrosis, acute renal failure Cerebral malaria: coma, death Pulmonary edema.	Similar to falciparum except vascular sequestration or cerebral involvement less common	Immune complex glomerulonephritis	Severe anemia, splenomegaly; Relapse due to liver infection (hyponozoite form infects hepatocytes)	Relapse due to latent liver infection (hyponozoite form infects hepatocytes)

	<i>Plasmodium falciparum</i>	<i>Plasmodium knowlesi</i>	<i>Plasmodium malariae</i>	<i>Plasmodium vivax</i>	<i>Plasmodium ovale</i>
Diagnosis	PCR; Rapid diagnostic tests; Blood smear: multiple parasites in single erythrocytes (high parasitemia), ring forms, crescent-shaped gametocytes	PCR required to differentiate from P. falciparum, P. malariae; Rapid diagnostic tests Blood smear: high parasitemia, ring and band forms	PCR; Rapid diagnostic tests Blood smear: band form trophozoites	PCR; Rapid diagnostic tests Blood smear: parasitized young RBCs, Schuffner's dots (RBC inclusions)	PCR; Rapid diagnostic tests Blood smear: parasitized young RBCs, Schuffner's dots; fimbriated infected RBCs
Prognosis	Requires emergent treatment. High risk of death; Grading: erythrocyte load	High risk of death (possibly higher than falciparum)	Good with treatment: may result in chronic, debilitating infection	Good with treatment	Good with treatment
Treatment	Artemisinin-based combination therapy (ACT, e.g., artemether-lumefantrine), Widespread chloroquine resistance	Chloroquine resistance	Chloroquine	Chloroquine if sensitive or ACT (followed by primaquine or tafenoquine to eradicate liver stage); Variable chloroquine resistance by region	Similar to P. vivax, except less risk of chloroquine resistance 



Protozoa: *Toxoplasma gondii*

- Causes toxoplasmosis
- Obligate intracellular parasite that infects all warm-blooded animals
- Worldwide, more prevalent in warm climates
- ¼ of world's population infected; 10% in U.S.
- Transmission from housecats, contaminated meat or shellfish
 - Cats eat infected rodents and shed oocysts in stool
- Vertical transmission causes congenital toxoplasmosis: triad of **chorioretinitis**, **hydrocephalus**, **intracranial calcifications**; less commonly **blueberry muffin rash**; prognosis variable to fatal
- Immune competent: usually asymptomatic, but may cause fever, lymphadenopathy, ocular disease; rarely myocarditis, hepatitis, disseminated disease
- Reactivation disease in immune suppressed (AIDS CD4 <100/ μ L, SOT and HSCT) results in cerebral toxoplasmosis, chorioretinitis and disseminated infection
- Diagnosis by serum antibody (elevated IgM in acute infection), ring-enhancing lesions on brain CT/MRI (appears similar to primary CNS lymphoma in AIDS)
- Treatment: pyrimethamine/sulfadiazine 4-6 weeks, clindamycin/pyrimethamine, atovaquone/azithromycin, for CNS disease trimethoprim/sulfamethoxazole

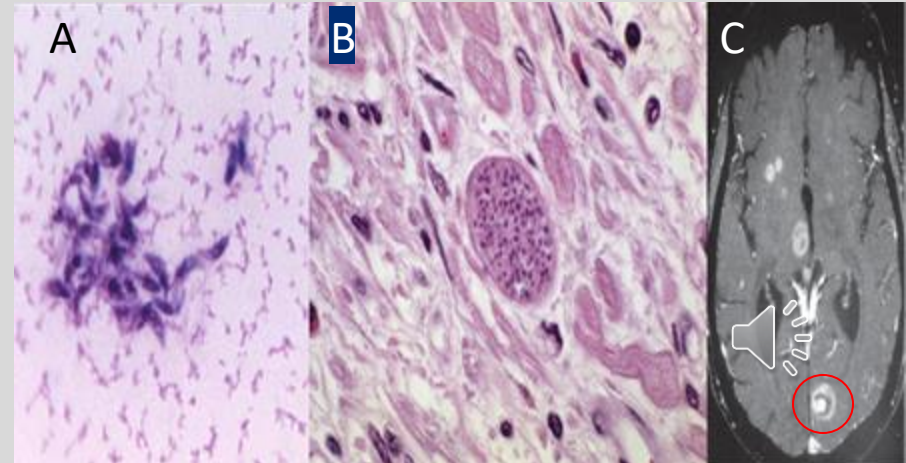


Figure: A. tachyzoites, B. tissue cyst, C. ring-enhancing lesion on MRI

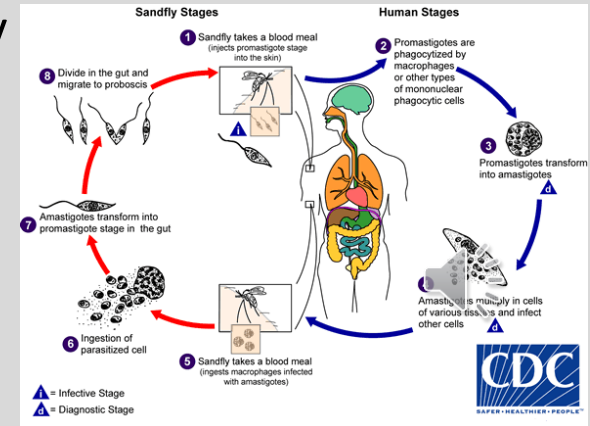
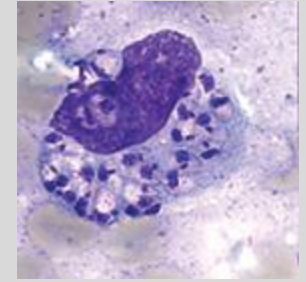
Protozoa: Leishmania

- Central and S. America, Africa, Middle East
- Blood-borne, obligate intracellular parasite transmitted by sandfly bite (>20 species infect humans)
 - Sandfly infected via blood meal from rodents, dogs, foxes
 - Adult sandfly injects human with promastigote stage
 - Motile promastigotes (flagella) infect macrophages, transform to amotile amastigotes (no flagella) which multiply
- Infection results in highly disfiguring ulcers or fatal organ involvement



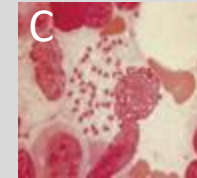
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Leishmaniasis

- **Cutaneous (A):** bite results in skin ulcer and eventual scar after about 1 year; may be diffuse in immune compromised; diagnosis by ulcer scraping showing macrophages filled with amastigotes;
- **Mucocutaneous (B):** **dermal ulcer that heals quickly** followed by mucous membrane ulceration, nasal septum erosion; may have secondary bacterial infection; primarily *L. braziliensis*; diagnosis by ulcer scraping
- **Visceral (Kala-azar, C):** fever, diarrhea, anemia, hepatosplenomegaly, reticuloendothelial spread; fatal if untreated; primarily *L. donovani* (although *L. donovani* can cause all three forms of disease); diagnosis by serology, liver or spleen biopsy
- **Post kala-azar dermal leishmaniasis (PKDL, D):** non-ulcer facial lesions occur during or after treatment; may be chronic
- **Treatment:** stilbogluconate (**contact CDC**), amphotericin B, others

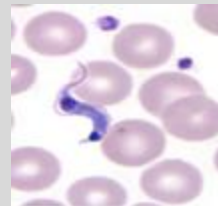


Protozoa: *Trypanosoma*

	<i>Trypanosoma cruzi</i>	<i>Trypanosoma brucei</i>
Geographic distribution	Rural areas from S. Texas to S. America; cases reported from CA, AZ, TN, LA, MO, MS, AR	T.b. gambiense: West Africa T.b. rhodesiense: East Africa
Disease	Chagas disease	African sleeping sickness
Transmission; reservoir	Reduviid (kissing) bug; dogs, armadillos, opossum	Tsetse fly; TbG: domestic animals (e.g., pigs, cattle) TbR: domestic and wild animals (e.g., bushback)
Clinical features	Most asymptomatic Fever, malaise, lymphadenopathy Chagoma: indurated, red rash at bite Romaña sign: periorbital edema and rash	Hard, painful, red ulcer at bite heals within 2 weeks. Early hemolymphatic stage: Winterbottom's sign (enlarged posterior cervical lymph node): followed slowly by other clinical features below:
Other clinical features	Chronic infection: Triad of irreversible megaesophagus, megacolon, dilated cardiomyopathy with or without RBBB	Systemic: slowly progressive relapsing fever, headache, dizziness, wasting Late CNS infection: lethargy, behavioral changes, ataxia, slurred speech, coma, death TbR: quicker progression of fever to death within weeks to months
Diagnosis	Clinical, serology Blood smear shows trypomastigote	Giemsa blood smear reveals trypomastigotes , lymph node aspirate, CSF analysis
Treatment	Nifurtimox or benznidazole	Pentamidine or suramin for systemic infection: Melarsoprol for CNS infection



Figure 5: Winterbottom's sign



Zika Virus



Zika virus

Geographic distribution	S. & Central America, Central & West Africa, SE Asia, FL, TX
Virus	Flaviviridae ; positive-sense, single-stranded, enveloped RNA virus
Transmission	Aedes aegypti or <i>A. albopictus</i> (tiger mosquito); Perinatal, gestational, sexual , blood transfusion, solid organ transplant
Clinical features	Asymptomatic in 80%; Fever, headache, arthralgia, myalgia, pruritic, maculopapular rash, conjunctivitis Congenital Zika syndrome : microcephaly in infants born to mothers infected in 1 st trimester
Complications	Guillain-Barre syndrome Co-infection with other Orthoflaviviruses transmitted by <i>Aedes</i> mosquito (dengue, chikungunya)
Diagnosis	Travel history PCR, ELISA, serology
Treatment	Supportive

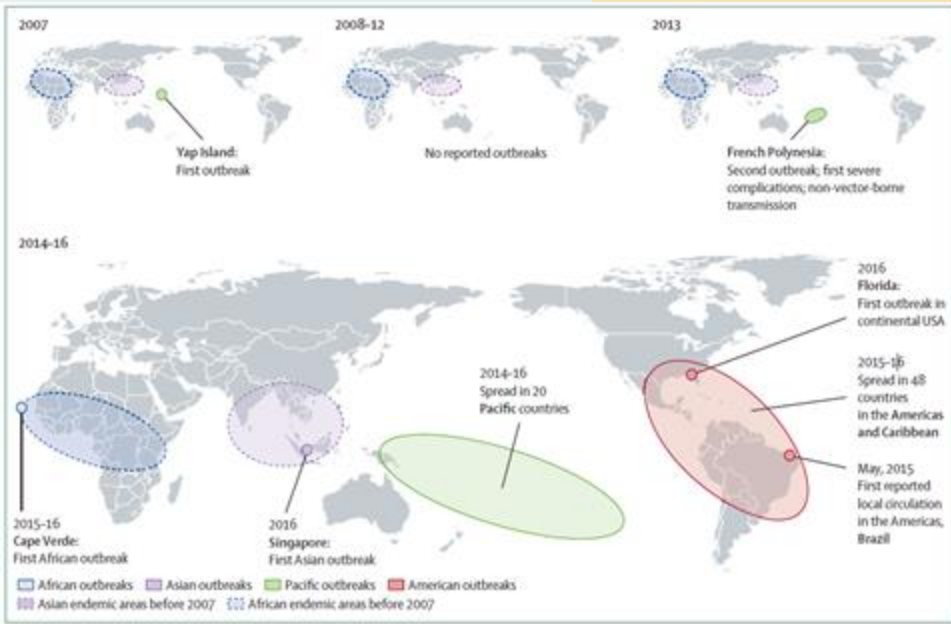
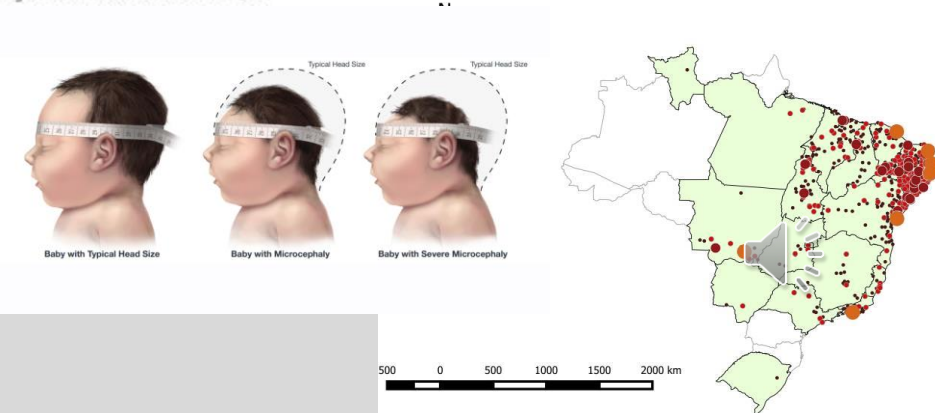
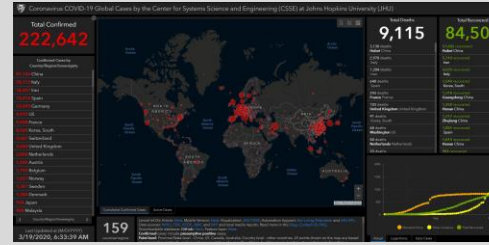


Figure 2: Zika virus outbreaks from 2007-16



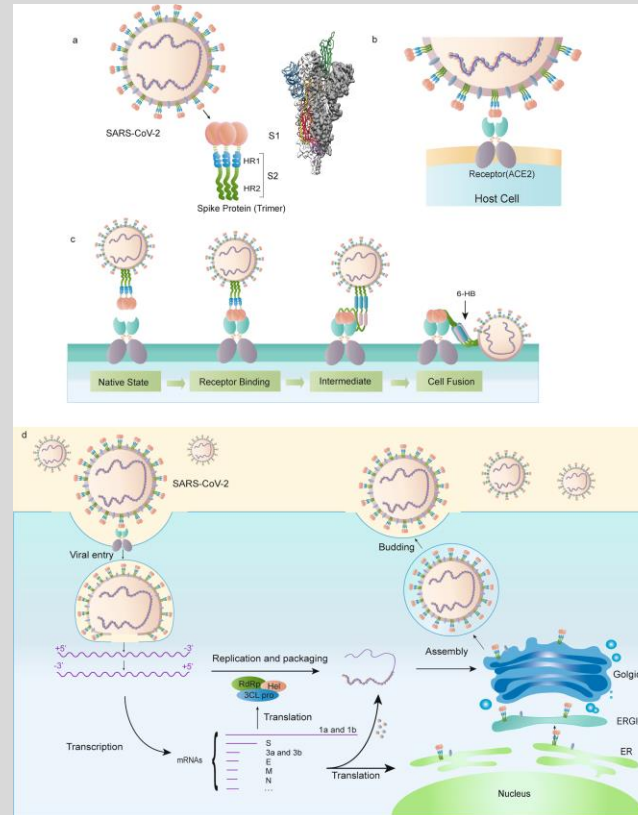
SARS-CoV-2

	SARS-CoV-2
Geographic distribution	Global
Virus	Coronaviridae; enveloped, positive-sense, single-stranded, RNA virus; spike protein binds ACE2R
Disease	COVID-19
Transmission	Aerosol, human-to-human, domesticated animals, zoo animals, mink, white-tailed deer
Incubation	3-7 days
Clinical features	Asymptomatic in 30%; Fever, cough, shortness of breath Viral pneumonia Severe illness in non-immune
Complications	Late inflammatory phase (“cytokine storm”) Co-infection with other respiratory viruses (influenza, RSV) Post-viral bacterial pneumonia Long-COVID in 40-50%
Diagnosis	Rapid antigen, PCR
Treatment	Nirmatrelvir/r, remdesivir, molnupiravir; Steroids for late inflammatory phase
Prevention	Vaccination



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Pathogens Discussed in this Lecture

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CLASS	PATHOGEN	DISEASE
Bacteria	<i>Bartonella henselae</i>	Cat Scratch Disease
	<i>Bartonella quintana</i>	Trench Fever
	<i>Bartonella bacilliformis</i>	Oroya Fever / Carrión Disease
	<i>Borrelia burgdorferi</i>	Lyme Disease
	<i>Borrelia miyamotoi</i>	Hard Tick Relapsing Fever
Protozoa	<i>Babesia microti</i>	Babesiosis
	<i>Plasmodium</i>	Malaria
	<i>Toxoplasma gondii</i>	Toxoplasmosis
	<i>Leishmania</i>	Leishmaniasis
	<i>Trypanosoma cruzi</i>	Chagas Disease
	<i>Trypanosoma brucei</i>	African Sleeping Sickness
Viruses	Zika	Zika Virus Disease
	SARS-CoV-2	COVID-19



Thank you for reviewing and good luck!

- Summary

1. The epidemiology, incidence and geographic distribution of vector-borne disease is evolving
2. Knowledge helps clinicians recognize and manage emerging pathogens
3. Expert consultation for emerging pathogens exists at the state (State Health Department) and federal (CDC) levels

- [Feedback](#)

